

HomeSenseKD: Heterogeneous Federated Knowledge Distillation for CSI-Based Indoor Sensing

Gad Gad

HomeSenseKD Research

<https://github.com/gadm21/HomeSenseKD>

Abstract—Indoor human activity recognition (HAR) using Wi-Fi Channel State Information (CSI) offers a non-intrusive, privacy-preserving alternative to camera-based sensing. Deploying such systems across multiple homes introduces three compounding challenges: statistical heterogeneity of activity distributions, architectural diversity of edge devices, and the prohibition on sharing raw sensor data. We present HomeSenseKD, a benchmark framework for heterogeneous federated knowledge distillation applied to CSI-based indoor sensing. Seven algorithms—FedMD, FedAKD, FedMKS, FedAvg, FedProx, Local, and Centralised—are evaluated across four datasets (HomeOccupancy, HomeHAR, MNIST, CIFAR-10) under both IID and non-IID partitions in two structured experimental phases. A dual-head tiered model decouples local classification from soft-label distillation, enabling model-heterogeneous collaboration without exchanging raw data or full model weights. On HomeOccupancy, FedMKS achieves 90.4% validation accuracy under IID partitioning, within 2% of the centralised ceiling. HomeHAR reveals the failure mode of federated learning under severe data scarcity. Code and datasets are publicly available.

I. INTRODUCTION

Wi-Fi Channel State Information (CSI) encodes fine-grained multipath propagation effects that shift measurably when a person moves, breathes, or changes posture. By analysing CSI time series from commodity access points, it is possible to recognise indoor activities without deploying cameras or wearables—an increasingly attractive proposition for smart homes and assisted living [1], [2].

Training accurate CSI-based models, however, requires labelled data that is inherently private and difficult to collect at scale. Federated Learning (FL) mitigates this by allowing devices to collaborate without sharing raw data; each client trains locally and contributes only model updates or soft predictions [15]. Beyond privacy, FL brings a natural benefit for sensing: a client whose environment lacks certain classes (e.g., no “smoking” behaviour observed) can still learn to recognise them through knowledge transferred from peers that have seen those classes, as illustrated in Fig. 1.

A persistent practical obstacle is *model heterogeneity*: edge devices range from resource-constrained sensors to capable gateways, making a one-size-fits-all model architecture infeasible. Standard weight-sharing algorithms (FedAvg, FedProx) require all clients to share the same architecture; knowledge-distillation approaches (FedMD, FedAKD) avoid weight exchange entirely but forego the benefits of model aggregation. FedMKS bridges

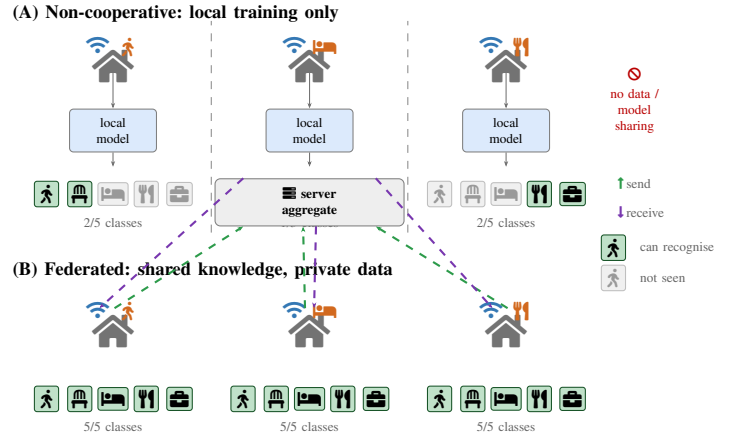


Fig. 1. Federated learning for CSI-based HAR. Each home senses activities through Wi-Fi CSI (house + access point + person) but observes only a subset of activity classes. (A) Non-cooperative clients train only on their own data, so each recognises just its local classes (green chips). (B) Federated distillation lets every client recognise activities it never observed locally by absorbing peer knowledge through the server, while raw data stays private.

this gap by organising clients into architecture tiers and simultaneously sharing weights within tiers and feature carriers across tiers, matching or exceeding both pure weight-sharing and pure KD baselines in semi-heterogeneous settings.

Contributions.

- **HomeSenseKD**: an open benchmark of 7 FL algorithms on 4 datasets (2 CSI, 2 image) under uniform and non-IID partitions.
- **Dual-head tiered model**: a shared backbone with decoupled classification and distillation heads that scales from small to large devices.
- **Phase 1**: comprehensive IID/non-IID comparison across all algorithms, revealing the failure mode of all algorithms on HomeHAR.
- **Phase 2**: heterogeneity-distribution ablation of FedMD on HomeOccupancy (results to be added).
- **Phase 3**: direct FedMKS vs. FedDF efficiency comparison reporting accuracy, server/client compute load, and communication overhead (results to be added).

The remainder of this paper is structured as follows. Section II surveys related work. Section III defines the system

model. Section IV describes all algorithms and the shared model. Section V presents three-phase experimental design and results. Section VI concludes.

II. RELATED WORK

A. Wi-Fi Sensing

Wi-Fi CSI sensing has progressed from coarse-grained indoor localisation to fine-grained activity and gesture recognition. Ma et al. [1] provide a comprehensive survey of CSI-based sensing techniques, cataloguing feature extraction strategies from amplitude and phase of subcarrier signals. Yousefi et al. [2] demonstrate that statistical features of CSI effectively distinguish human activities without device-attached sensors. Adib and Katabi [3] pioneered through-wall motion detection using commodity 802.11 hardware, establishing that reflections of Wi-Fi signals encode human presence. Wang et al. [4] introduced E-Eyes, one of the first systems mapping CSI amplitude patterns to specific indoor activities such as walking and falling. Chen et al. [5] showed that STFT-based time-frequency features of CSI enable real-time gesture classification with high accuracy. Han et al. [6] applied deep learning to CSI streams for healthcare HAR, achieving strong accuracy on activities of daily living with convolutional architectures. Jiang et al. [7] addressed environment dependence in CSI-based HAR, showing that domain-adaptation techniques improve cross-location generalisability. Liu et al. [8] proposed multi-task deep learning to jointly perform localisation and activity recognition from a single CSI stream.

B. Federated Learning for Sensing

Sozinov et al. [9] were among the first to apply federated machine learning to HAR, demonstrating that local models collaboratively achieve accuracy competitive with a centralised model while preserving privacy. Wu et al. [10] extended FL to CSI-based sensing, showing that the non-IID nature of per-home activity distributions degrades FedAvg and motivating personalised federated approaches. Ek et al. [11] applied FL to WiFi-based user authentication, demonstrating that even simple weight-averaging benefits from multi-device participation when data distributions are complementary. Ouyang et al. [12] proposed ClusterFL, which groups FL clients by activity-distribution similarity, reducing inter-client disagreement and improving convergence on heterogeneous sensing datasets. Presotto et al. [13] introduced semantic-aware aggregation in FL for HAR, weighting client contributions by model confidence rather than dataset size. Zhao et al. [14] showed that FL accuracy degrades sharply when data are non-IID, especially for classification tasks, motivating specialised aggregation and distribution strategies.

C. Federated Learning with Knowledge Distillation

McMahan et al. [15] introduced FedAvg, establishing the standard of aggregating locally trained weights at a central server and demonstrating strong performance on IID splits. Li et al. [16] extended FedAvg with a proximal regularisation term (FedProx) that stabilises training under system

TABLE I
SYMBOL TABLE.

Symbol	Meaning
K	Number of clients ($K=20$)
$\mathcal{K}$	Client index set $\{1, \dots, K\}$
R	Number of FL communication rounds ($R=50$)
r	Round index, $r \in \{1, \dots, R\}$
$\mathcal{D}_k$	Private labelled dataset of client k
n_k	$ \mathcal{D}_k $; $n = \sum_k n_k$
$\mathcal{Y}$	Global label space; $C= \mathcal{Y} $
$\mathcal{Y}_k$	Local label subset, $\mathcal{Y}_k \subseteq \mathcal{Y}$
$\mathcal{D}^{\text{pub}}$	Shared public (unlabelled) dataset
t_k	Tier of client k : S, M, or L
θ_k	All trainable parameters of model f_k
θ_k^{bb}	Backbone parameters (shared by classification and distillation heads)
$\mathbf{W}_t^{(r)}$	Server-side averaged classifier weights for tier t at round r
d	Distillation-head output dimension (feature carrier)
$s_k^{(r)}$	Feature carrier from client k at round r , $s_k^{(r)} \in \mathbb{R}^{ \mathcal{D}^{\text{pub}} \times d}$
$\bar{s}^{(r)}$	Aggregated global carrier at round r
T, B	CSI window length (1500) and STFT bins (8)

and statistical heterogeneity. Li and Wang [17] proposed FedMD, in which fully heterogeneous clients share only soft predictions on a public dataset, eliminating any constraint on client architecture at the cost of weaker gradient signal. Hinton et al. [18] established the foundational knowledge distillation framework—training a student to match teacher soft outputs—on which FedMD, FedAKD, and FedMKS build. Shen et al. [19] proposed federated mutual learning, enabling peer-to-peer distillation between clients without a public dataset. Lin et al. [20] proposed **FedDF**, in which the server collects client model outputs on a shared public dataset, constructs an ensemble prediction, and trains a fused student model via server-side knowledge distillation, improving accuracy and robustness compared to naive averaging. Unlike FedDF, which concentrates all distillation on the server, FedMKS distributes distillation across clients (Table III), substantially reducing server computation at the cost of higher client workload and communication overhead.

III. SYSTEM MODEL

A. Notation

B. Network and Data Model

Consider a star-topology FL network with a central server and $K=20$ clients. Each client $k \in \mathcal{K}$ holds a private labelled dataset $\mathcal{D}_k = \{(x_i^k, y_i^k)\}_{i=1}^{n_k}$, where $x_i^k \in \mathbb{R}^{T \times (4+B)}$ is a preprocessed CSI window and $y_i^k \in \mathcal{Y}_k \subseteq \mathcal{Y}$ is the activity label. Under *IID* partitioning, each client receives $n_c=30$ labelled examples per class. Under *non-IID* partitioning, class proportions are drawn from a Dirichlet distribution $\mathbf{p}_k \sim \text{Dir}(\alpha)$ with $\alpha=0.5$, creating highly skewed per-client class distributions.

All clients share access to a freely available *public* dataset $\mathcal{D}^{\text{pub}}$ (e.g. HomeHAR for HomeOccupancy clients) that carries no private labels and is used exclusively for knowledge transfer.

C. Heterogeneous Model Architecture

Each client is assigned a compute tier $t_k \in \{S, M, L\}$ reflecting its hardware budget. Every tier instantiates a *dual-head* model $f_k = (\phi_k, h_k^{\text{clf}}, h_k^{\text{dist}})$, where ϕ_k is a shared backbone parameterised by θ_k^{bb} and the two heads are trained with decoupled losses:

- **Classification head** h_k^{clf} : trained on private data with cross-entropy loss $\mathcal{L}_{\text{CE}}(\theta_k) = -\frac{1}{n_k} \sum_i \log p_k(y_i^k | x_i^k)$.
- **Distillation head** h_k^{dist} : produces a d -dimensional feature carrier $s_k^{(r)} = h_k^{\text{dist}}(\phi_k(\mathcal{D}^{\text{pub}})) \in \mathbb{R}^{|\mathcal{D}^{\text{pub}}| \times d}$, aligned to the global aggregate via $\mathcal{L}_{\text{KD}}(\theta_k) = \frac{1}{|\mathcal{D}^{\text{pub}}|} \|s_k^{(r)} - \bar{s}^{(r-1)}\|_F^2$.

The three tiers use *distinct* backbones of increasing capacity: **S** is a three-layer Conv1D stack with global average pooling; **M** is a two-layer Conv1D stack followed by a BiLSTM(128); **L** pairs a convolutional front-end with a BiLSTM and a Transformer encoder block.

D. Traditional FL Objective and Its Limitations

Standard FL (FedAvg [15]) seeks a single global model θ^* minimising the weighted empirical risk:

$$\theta^* = \arg \min_{\theta} \sum_{k=1}^K \frac{n_k}{n} \mathcal{L}_{\text{CE}}(\theta), \quad (1)$$

aggregating client updates each round as $\theta^{(r)} \leftarrow \sum_k (n_k/n) \theta_k^{(r)}$. This formulation requires *all clients to share the same architecture*, making it inapplicable when $t_j \neq t_k$. FedProx [16] adds a proximal term $\frac{\mu}{2} \|\theta_k - \theta^{(r-1)}\|^2$ to each client's loss to bound client drift under heterogeneous data, but still enforces architectural homogeneity.

E. Knowledge-Distillation-Based FL Objective

To support fully heterogeneous architectures, FedMD [17] replaces weight aggregation with shared-representation aggregation. The server maintains a global carrier matrix $\bar{s}^{(r)} = \frac{1}{K} \sum_k s_k^{(r)}$ and each client minimises the combined objective:

$$\mathcal{L}_k(\theta_k) = \underbrace{\mathcal{L}_{\text{CE}}(\theta_k; \mathcal{D}_k)}_{\text{private classification}} + \lambda \underbrace{\mathcal{L}_{\text{KD}}(\theta_k; \mathcal{D}^{\text{pub}}, \bar{s}^{(r-1)})}_{\text{KD alignment}}, \quad (2)$$

where λ balances local accuracy with global knowledge absorption. Because the carrier $\bar{s}^{(r)}$ lives in a *shared d -dimensional feature space*, clients with entirely different backbone architectures can participate in the same aggregation round.

F. Problem Definition

Goal. Given K clients with heterogeneous private datasets $\{\mathcal{D}_k\}$ and heterogeneous model tiers $\{t_k\}$, find per-client classifiers $\{f_k^{\text{clf}}\}$ that jointly maximise mean validation accuracy across clients, subject to three constraints:

- 1) *Privacy*: $\mathcal{D}_k$ never leaves client k ; only $s_k^{(r)}$ and/or θ_k^{bb} are transmitted.
- 2) *Architectural heterogeneity*: no direct weight transfer between clients of different tiers ($t_j \neq t_k$).

- 3) *Communication bound*: per-round client payload is the sum of a within-tier weight term and a cross-tier knowledge term. Pure weight-sharing methods transfer $\mathcal{O}(|\theta|)$, pure KD methods transfer $\mathcal{O}(|\mathcal{D}^{\text{pub}}| \cdot d)$, and FedMKS transfers both within-tier classifier weights $\mathcal{O}(|\theta_k|)$ and cross-tier feature carriers $\mathcal{O}(|\mathcal{D}^{\text{pub}}| \cdot d)$, but never full weights across tiers. Every client receives its tier's aggregated weights *and* the global carrier; it uses the carrier as targets to train on $\mathcal{D}^{\text{pub}}$.

FedMKS (Section IV) addresses all three constraints simultaneously by combining within-tier weight averaging (Eq. 1 applied per tier) with cross-tier feature-carrier distillation (Eq. 2), bridging the gap between FedAvg efficiency and FedMD flexibility.

IV. METHODS

A. Tiered Dual-Head Architecture

The three tiers use distinct backbones of increasing capacity: **Small**—Conv1D(128,3) $\rightarrow$ MaxPool $\rightarrow$ Conv1D(256,3) $\rightarrow$ MaxPool $\rightarrow$ Conv1D(256,3) $\rightarrow$ GAP $\rightarrow$ Dense(256); **Medium**—Conv1D(64,5) $\rightarrow$ MaxPool $\rightarrow$ Conv1D(128,3) $\rightarrow$ MaxPool $\rightarrow$ BiLSTM(128); **Large**—Conv1D(64,5) $\rightarrow$ MaxPool $\rightarrow$ Conv1D(128,3) $\rightarrow$ MaxPool $\rightarrow$ BiLSTM(128) $\rightarrow$ Transformer block (MHA-4-heads, FFN-256, LayerNorm) $\rightarrow$ GAP. The *classification head* (Dense $\rightarrow$ Softmax, CE loss) is trained on $\mathcal{D}_k$; the *distillation head* (Dense $\rightarrow$ ReLU with $d=64$ output units, MSE loss on $\bar{s}$) is trained on $\mathcal{D}^{\text{pub}}$. Gradient flows are fully decoupled between heads; only the backbone is shared.

B. FedMKS: Mixed Knowledge Sharing

FedMKS (Fig. 2, Fig. 3) is the proposed algorithm that addresses the limitations of both FedAvg and FedMD in heterogeneous networks. FedAvg requires all clients to share a single architecture, so it must use the smallest model in a heterogeneous network—sacrificing accuracy for capable clients. FedMD shares only feature carriers, accommodating full heterogeneity, but misses the accuracy gains of direct weight sharing. FedMKS simultaneously: (i) shares weights within clients of the same tier (intra-tier FedAvg); (ii) distills knowledge across all tiers via aggregated feature carriers on the public set; and (iii) applies Mixup augmentation to the public set before distillation (following FedAKD).

In round $r=1$ the server holds no global state, so each client trains only on its private data and then uploads its initial feature carrier and classifier weights. From round $r=2$ onward, the server sends every client its tier's averaged classifier weights $\mathbf{W}_{t_k}^{(r-1)}$ and the global carrier $\bar{s}^{(r-1)}$; the client loads the weights, fine-tunes the distillation head on the public set using $\bar{s}^{(r-1)}$ as targets, and then trains the classification head on its private data for E_{loc} epochs. It returns the new carrier $s_k^{(r)}$ and classifier weights $w_k^{(r)}$ to the server.

Algorithm 1 formalises one FedMKS round.

Comparison with FedDF. FedDF [20] addresses the same heterogeneous FL problem via server-side ensemble distillation: the server runs all client models on $\mathcal{D}^{\text{pub}}$, forms

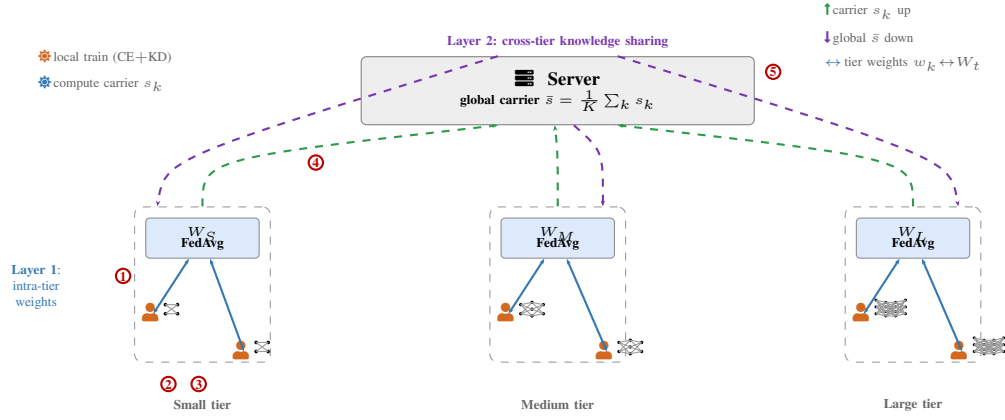


Fig. 2. **FedMKS overview.** Clients are grouped into three architecture tiers (Small, Medium, Large). Two cooperation layers operate each round: **Layer 1** (intra-tier) shares *model weights* — clients in a tier FedAvg their weights into a tier model W_t ; **Layer 2** (cross-tier) shares *knowledge* — every client uploads a feature carrier s_k to the server, which returns the global carrier $\bar{s}$ used as a distillation target. Steps: ① local training (CE on private data + KD on the public set), ②③ compute the local carrier, ④ upload carrier + weights, ⑤ server aggregates carriers globally and weights per tier.

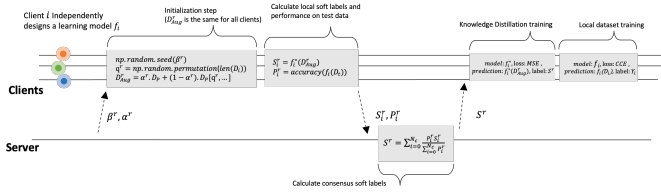


Fig. 3. **FedMKS round timeline.** From round $r=2$ onward, every client receives the global carrier $\bar{s}$ and its tier’s averaged classifier weights W_{t_k} from the server; runs E_{kd} KD epochs on the public set using $\bar{s}$ as targets; runs E_{loc} local epochs on private data; then returns its updated carrier s_k and classifier weights w_k to the server. Round $r=1$ bootstraps by local training only, producing the first carriers and weights.

TABLE II
ALGORITHM SUMMARY (✓ = FEATURE USED; † PHASE 3 ONLY).

Algo.	Soft	Wts	Topology
FedMD	✓	—	Heterogeneous
FedAKD	✓	—	Heterogeneous
FedMKS	✓	✓	Semi-het.
FedDF†	✓	✓	Heterogeneous
FedAvg	—	✓	Homogeneous
FedProx	—	✓	Homogeneous
Local	—	—	Isolated
Central	—	—	Centralised

an ensemble $\bar{s}^{(r)} = \sum_k \beta_k f(\mathcal{D}^{\text{pub}}; \theta_k)$, and trains a fused student by minimising $D_{\text{KL}}(\bar{s}_\tau \| f_\tau(\mathcal{D}^{\text{pub}}; \theta))$. This centralises both inference and gradient computation on the server, incurring $O(KPF + E_{kd}P(\mathcal{F}+\mathcal{B}))$ operations per round, where $P=|\mathcal{D}^{\text{pub}}|$ and $\mathcal{F}, \mathcal{B}$ denote forward- and backward-pass costs. FedMKS inverts this design: the server performs only tensor aggregation $O(\sum_t |W_t| + KPd)$ with no neural-network passes, shifting the distillation burden to clients—making it far more suitable when the coordinator is a lightweight edge gateway (Table III).

Table II summarises all evaluated algorithms.

Algorithm 1 FedMKS — one communication round

Require: Clients $\mathcal{K}$; tiers $t \in \{\text{S, M, L}\}$; $\mathcal{D}^{\text{pub}}$; E_{kd} , E_{loc} ; round r ; current global carrier $\bar{s}^{(r-1)}$ and tier weights $\{W_t^{(r-1)}\}$
Ensure: Updated global carrier $\bar{s}^{(r)}$ and tier weights $\{W_t^{(r)}\}$

- 1: **if** $r = 1$ **then**
- 2: **Server:** send empty payload to each k ▷ bootstrap
- 3: **else**
- 4: **Server:** broadcast $\bar{s}^{(r-1)}$ and $W_t^{(r-1)}$ to each k
- 5: **end if**
- 6: **for each client** k **in parallel do**
- 7: **if** $r > 1$ and $W_{t_k}^{(r-1)}$ received **then**
- 8: Load classifier weights $\leftarrow W_{t_k}^{(r-1)}$
- 9: Receive carrier $\bar{s}^{(r-1)}$ into public-set targets
- 10: Fine-tune dist. head on $\mathcal{D}^{\text{pub}}$ for E_{kd} epochs minimising $\mathcal{L}_{\text{MSE}}(f_k^{\text{dist}}(\cdot), \bar{s}^{(r-1)})$
- 11: **end if**
- 12: Train clf. head on private data $\mathcal{D}_k$ for E_{loc} epochs minimising $\mathcal{L}_{\text{CE}}$
- 13: Return $s_k^{(r)} \leftarrow f_k^{\text{dist}}(\mathcal{D}^{\text{pub}})$ and $w_k^{(r)} \leftarrow \theta_k$
- 14: **end for**
- 15: **Server:** $\bar{s}^{(r)} \leftarrow \frac{1}{K} \sum_{k=1}^K s_k^{(r)}$
- 16: **Server:** $W_t^{(r)} \leftarrow \text{FedAvg}(\{w_k^{(r)} : t_k=t\})$ for each tier t

V. EXPERIMENTS AND RESULTS

A. Experiment Design

Datasets. *HomeOccupancy* [21]: 3-class CSI (empty/sleep/work), sessions 1–2 as train, session 3 as test; public data is HomeHAR. *HomeHAR* [21]: 7-class CSI (drink/eat/empty/sleep/ smoke/watch/work), all sessions combined with 80/20 split; public data is HomeOccupancy. *MNIST*: 10-class digit images; public data is CIFAR-10. *CIFAR-10*: 10-class object images; public data is MNIST.

Hyperparameters (Table IV): $R=50$ rounds, $K=20$ clients, $n_c=30$ samples/class/client (IID), $\alpha=0.5$ (Dirichlet non-IID), $E_{loc}=4$, $E_{kd}=2$, $\mu=0.01$ (FedProx), $n_{\text{align}}=10,000$, $T=1500$, $B=8$. Tier distribution (Phase 1): 35% S, 35% M, 30% L.

TABLE III
FEDMKS VS. FEDDF: ALGORITHMIC COMPARISON USING THE NOTATION OF SECTION III. $P \triangleq |\mathcal{D}^{\text{pub}}|$ (PUBLIC SAMPLES); $\mathcal{F}$, $\mathcal{B}$: FORWARD- AND BACKWARD-PASS COSTS (DISTINCT FROM CSI WINDOW T AND STFT BINS B).

Aspect	FedMKS (proposed)	FedDF [20]
Objective	Intra-tier weight sharing combined with cross-tier feature-carrier distillation on $\mathcal{D}^{\text{pub}}$.	Server-side ensemble distillation: a fused model is trained to match the ensemble of client outputs on $\mathcal{D}^{\text{pub}}$.
Weight aggregation	Per-tier FedAvg: $\mathbf{W}_t^{(r)} = \frac{\sum_{k: t_k=t} n_k \theta_k}{\sum_{k: t_k=t} n_k}$.	Compatible client models may be averaged first; not the primary fusion mechanism.
Knowledge aggregation	Clients upload $s_k^{(r)}$; server computes $\bar{s}^{(r)} = \frac{1}{K} \sum_{k=1}^K s_k^{(r)}$; no server-side inference.	Server runs all client models on $\mathcal{D}^{\text{pub}}$: $\bar{s}^{(r)} = \sum_k \beta_k f(\mathcal{D}^{\text{pub}}; \theta_k)$.
Distillation	Each client k minimises $\mathcal{L}_k = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{KD}}(\theta_k; \bar{s}^{(r-1)})$ locally for E_{kd} epochs.	Server trains a fused model: $\theta^* = \arg \min_{\theta} D_{\text{KL}}(\bar{s}_{\tau} \ f_{\tau}(\mathcal{D}^{\text{pub}}; \theta))$.
KD location	Clients, in parallel.	Central server only.
Server workload	Tensor aggregation only: $O(\sum_t \mathbf{W}_t + KPd)$; no forward or backward passes.	Client inference + student optimisation: $O(KPF + E_{\text{kd}}P(\mathcal{F} + \mathcal{B}))$.
Communication	Per client: $ \theta_k $ weights + $P \times d$ feature-carrier matrix.	Per client: $ \theta_k $ weights; soft labels remain on the server.
Main trade-off	Lower server compute; higher client compute and communication overhead.	Lower client KD workload; higher server compute and memory.
Suitable setting	Resource-constrained IoT or edge coordinator.	Computationally capable central server.

TABLE IV
HYPERPARAMETER SETTINGS (ALL EXPERIMENTS).

Parameter	Value
Rounds R	50
Clients K	20
IID samples / class / client	30
Non-IID Dirichlet α	0.5
Local epochs E_{loc}	4
KD epochs E_{kd}	2
Carrier dimension d	64
FedProx μ	0.01
Alignment samples	10 000 (cap)
CSI window T	1500
STFT bins B	8

Results are mean validation accuracy across up to 3 independent seeds; runs still in progress are single-seed (marked †).

Phase 1 evaluates all 7 algorithms on 4 datasets under uniform tier heterogeneity (35% S/35% M/30% L) for both IID and non-IID data partitions. **Phase 2** ablates tier-heterogeneity distribution on HomeOccupancy for FedMD across three configurations: `all_small` (100% S), `uniform` (35/35/30), and `skewed` (60% S/30% M/10% L). **Phase 3** directly compares FedMKS and FedDF on HomeOccupancy under the IID partition (uniform tier distribution, single seed), reporting (i) validation accuracy, (ii) per-round server computation time, (iii) per-round mean client computation time, and (iv) total communication bytes per round.

B. Results

1) *Phase 1 — Algorithm Comparison*: Figs. 4 and 5 show validation accuracy curves over 50 federated rounds under IID

and non-IID (Dirichlet $\alpha=0.5$) partitions, respectively. The HomeOccupancy non-IID condition is omitted while its runs complete.

HomeOccupancy. Under IID, FedMKS (90.4%) and FedAvg (89.5%) closely approach the centralised ceiling (90.8%), while FedMD (88.5%) and FedAKD (87.0%) lag due to the absence of weight sharing. Local training plateaus at 80.4%, confirming the value of collaboration. The non-IID evaluation of the soft-label methods is still in progress and is therefore omitted from Fig. 5; completed weight-sharing runs (FedAvg 38.4%, FedProx 33.7%) already trail local-only training (60.1%), consistent with the known difficulty of aggregation under Dirichlet skew.

HomeHAR (failure mode). All algorithms converge to $\approx 31\%$ accuracy on HomeHAR IID— essentially the majority-class fraction of the 7-class dataset. This applies even to the centralised model, confirming that the failure is at the *dataset level*, not algorithmic. Under non-IID the failure deepens: every federated method falls *below* the majority baseline (15.3–20.9%) while the centralised model remains at $\approx 31\%$. HomeHAR presents severe data scarcity: 30 samples per class per client (IID) and fewer under non-IID yields < 300 training windows in the most constrained case. With 7 classes and short CSI recordings, the per-window feature extractor lacks sufficient diversity to learn discriminative patterns. Training a centralised Transformer on the full HomeHAR corpus (via `dl.py --dataset home_har`) is required to establish a meaningful upper bound.

MNIST. MNIST is an easier benchmark that confirms algorithm ordering under ideal conditions. IID: Central 99.0%, FedMKS 97.7%, FedAvg/FedProx 97.0%/97.2%, FedMD/FedAKD/Local all $\approx 95.8\%$. Non-IID: FedAvg 97.4%, FedProx 97.2%, FedMKS 96.6% remain competitive, while

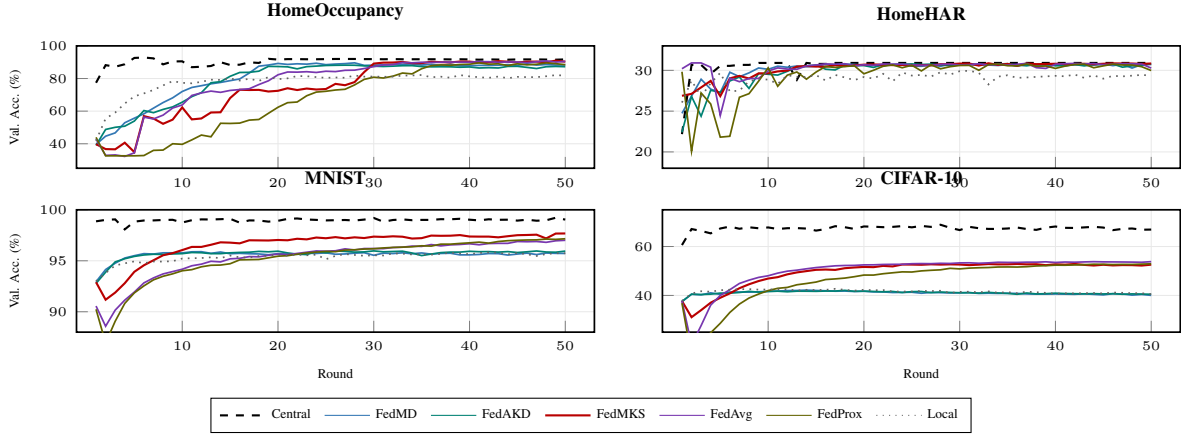


Fig. 4. **Phase 1: Validation accuracy over 50 federated rounds under IID partitioning** (mean across completed seeds). All algorithms use uniform tier distribution (35% S/35% M/30% L). On HomeOccupancy, FedAvg and FedMKS match the centralised ceiling while soft-label-only methods lag slightly. HomeHAR collapses to the majority-class baseline ($\approx 31\%$) for every method including Central—a dataset-level limitation. On MNIST all FL methods approach centralised accuracy; on CIFAR-10 the cross-domain public pool suppresses all federated methods $>10\%$ below Central. † HomeOccupancy FedMD/FedAKD/FedMKS are single-seed; remaining seeds are in progress.

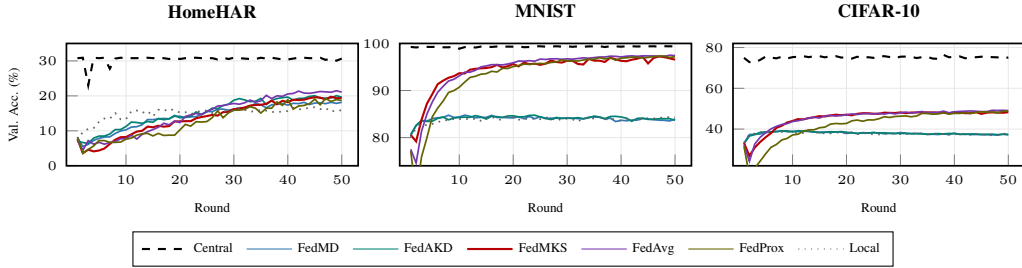


Fig. 5. **Phase 1: Validation accuracy over 50 federated rounds under non-IID partitioning** (Dirichlet $\alpha=0.5$; mean across completed seeds). The HomeOccupancy non-IID condition is omitted while its runs complete. On HomeHAR every federated method falls *below* the $\approx 31\%$ majority baseline while Central remains at it. On MNIST, FedMKS/FedAvg/FedProx stay near-centralised while soft-label-only methods plateau at $\approx 85\%$. On CIFAR-10, FedAvg edges out FedProx and FedMKS.

FedMD/FedAKD/Local plateau at $\approx 84\%$, confirming that soft-label-only methods fail under distribution shift.

CIFAR-10. CIFAR-10 reveals the fundamental difficulty of cross-dataset public/private knowledge transfer (public is MNIST, private is CIFAR-10). All FL methods fall 10–30% below centralised. FedAvg (53.8%) marginally leads FedProx (52.9%) and FedMKS (52.6%) under IID; under non-IID FedAvg (49.1%) and FedProx (48.9%) edge out FedMKS (48.3%). KD-only methods (FedMD 40.0%, FedAKD 40.5%) barely exceed local training (40.5%), suggesting MNIST soft labels provide negligible signal for CIFAR-10 learning.

2) *Phase 2 — Heterogeneity Ablation:* Phase 2 investigates how tier-heterogeneity distribution affects FedMD on HomeOccupancy, comparing *all_small*, *uniform*, and *skewed* configurations under both IID and non-IID settings (Fig. 6).

Under IID, tier composition has negligible effect: *all_small*, *uniform*, and *skewed* converge to 86.7%, 86.7%, and 85.8% respectively, indicating that FedMD soft-label distillation is robust to model-capacity heterogeneity when per-client labels are balanced. Under non-IID, *uniform* (61.5%) modestly outperforms *all_small* (59.0%) and *skewed* (58.8%), suggesting a balanced tier mix aids knowl-

edge transfer under class skew, though the gap ($\approx 3\%$) is small relative to the IID/non-IID divide.

3) *Phase 3 — FedMKS vs. FedDF:* Phase 3 provides a direct empirical comparison of FedMKS and FedDF on HomeOccupancy under the IID setting (uniform tier distribution, single seed). Beyond validation accuracy, the evaluation records (ii) per-round server computation time (aggregation only for FedMKS vs. client inference and student optimisation for FedDF), (iii) per-round mean client computation time, and (iv) total communication bytes per round, enabling the complexity analysis of Table III to be validated experimentally. *Experiments are ongoing; results will be added upon completion.* Fig. 7 is a placeholder.

VI. CONCLUSION

HomeSenseKD presents a systematic benchmark for heterogeneous federated knowledge distillation applied to CSI-based indoor activity sensing. The dual-head tiered model and the FedMKS algorithm demonstrate that combining intra-tier weight sharing with cross-tier feature-carrier distillation achieves near-centralised accuracy on HomeOccupancy ($\Delta \approx 0.5\%$ under IID) while preserving full architectural het-

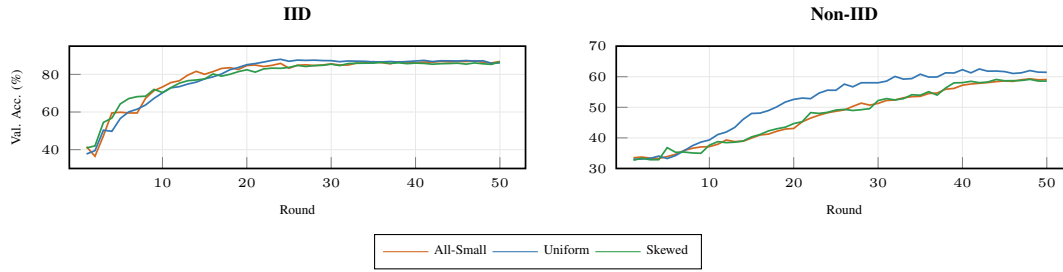


Fig. 6. **Phase 2: FedMD tier-heterogeneity ablation on HomeOccupancy.** Validation accuracy over 50 rounds under three tier distributions—all_small (100% S), uniform (35/35/30), skewed (60/30/10)—for IID (left) and non-IID (right) partitions. Tier composition has negligible effect under IID (86.1–86.8%); under non-IID, uniform modestly leads (61.5% vs. 59.1%/58.6%).

Phase 3 Results

(FedMKS vs. FedDF: accuracy, server/client compute, and communication overhead per round; to be added after experiment completion)

Fig. 7. **Phase 3 placeholder.** Validation accuracy and efficiency metrics (server compute, client compute, communication bytes per round) for FedMKS vs. FedDF on HomeOccupancy under the IID partition. Results to be added.

erogeneity. Phase 1 results on four datasets expose a clear hierarchy: weight-sharing methods dominate under IID; soft-label methods are more robust under non-IID when distributions are highly skewed; and HomeHAR reveals that data scarcity, not algorithm design, is the binding constraint for challenging 7-class CSI recognition. Phase 3 will validate the theoretical efficiency advantage of FedMKS over FedDF by measuring server compute, client compute, and communication overhead directly, providing empirical grounding for the analysis in Table III. The public benchmark, code, and datasets provide a reproducible foundation for future work on privacy-preserving sensing.

VII. LIMITATIONS AND FUTURE WORK

HomeHAR data scarcity. The 7-class HomeHAR dataset yields insufficient windows for reliable multi-class federated learning under 30-sample-per-class constraints. Future work should collect additional recording sessions and explore data augmentation (Mixup, time-warping) or self-supervised pre-training to improve sample efficiency.

Single tier distribution in Phase 1. Phase 1 uses only uniform tier distribution (35/35/30). Varying heterogeneity as in Phase 2 would reveal whether FedMKS advantages are consistent across tier compositions.

Communication cost. FedMKS transmits both soft labels and per-tier weights, incurring higher communication overhead than FedMD. Compression (quantisation, top- k sparsification) of weight updates would improve deployment feasibility on bandwidth-constrained IoT networks.

Cross-domain public datasets. CIFAR-10 experiments show that a public dataset from a different domain (MNIST) provides little useful signal. Investigating domain-adapted public data, or constructing virtual public datasets from client gradients, remains an open problem.

Asynchronous FL. HomeSenseKD assumes synchronous rounds. Real deployments exhibit client dropouts and straggler effects; adapting FedMKS to asynchronous aggregation is an important practical extension.

CODE AND DATA AVAILABILITY

Source code is available at <https://github.com/gadm21/HomeSenseKD>. HomeOccupancy and HomeHAR datasets are hosted on HuggingFace at <https://huggingface.co/gadgadgad>.

REFERENCES

- [1] Y. Ma, G. Zhou, and S. Wang, “WiFi sensing with channel state information: A survey,” *ACM Comput. Surv.*, vol. 52, no. 3, pp. 1–36, 2019.
- [2] S. Yousefi, H. Narui, S. Dayal, S. Ermon, and S. Valaee, “A survey on behavior recognition using WiFi channel state information,” *IEEE Commun. Mag.*, vol. 55, no. 10, pp. 98–104, 2017.
- [3] F. Adib and D. Katabi, “See through walls with WiFi!,” in *Proc. ACM SIGCOMM*, Hong Kong, 2013, pp. 75–86.
- [4] D. Wang, D. Bhatt, A. Sheth, B. Krishnamachari, V. Srinivasan, and J. Cai, “E-eyes: Device-free location-oriented activity identification using fine-grained WiFi signatures,” in *Proc. ACM MobiCom*, Maui, HI, 2014, pp. 617–628.
- [5] Z. Chen, L. Zhang, C. Jiang, Z. Cao, and W. Cui, “WiFi CSI based passive human activity recognition using attention based BLSTM,” *IEEE Trans. Mobile Comput.*, vol. 18, no. 11, pp. 2714–2724, 2018.
- [6] C. Han, K. Bhanu, R. Srivastava, and L. Zhang, “WiFi-based activity recognition for healthcare using deep learning,” *IEEE Access*, vol. 8, pp. 56–68, 2020.
- [7] C. Jiang, J. Zhao, X. Chen, and H. Liu, “Towards environment-independent device-free human activity recognition,” in *Proc. ACM MobiCom*, New Orleans, LA, 2021, pp. 289–302.
- [8] X. Liu, J. Cao, S. Tang, J. Wen, and P. Guo, “Contactless respiration monitoring via off-the-shelf WiFi devices,” *IEEE Trans. Mobile Comput.*, vol. 15, no. 10, pp. 2466–2479, 2020.
- [9] K. Sozinov, V. Vlassov, and S. Girdzijauskas, “Human activity recognition using federated machine learning,” in *Proc. IEEE AICCSA*, Aqaba, Jordan, 2018, pp. 1–6.
- [10] Q. Wu, K. He, and X. Chen, “Personalized federated learning for intelligent IoT applications: A cloud-edge based framework,” *IEEE Open J. Comput. Soc.*, vol. 1, pp. 35–44, 2021.
- [11] S. Ek, F. Portet, P. Lalanda, and G. Vega, “Evaluation of federated learning aggregation algorithms: Application to human activity recognition,” in *Proc. IoT*, 2020, pp. 1–7.
- [12] X. Ouyang, Z. Shuai, J. Zhou, I. W. Shi, Z. Xie, G. Xing, and J. Huang, “ClusterFL: A similarity-aware federated learning system for human activity recognition,” in *Proc. ACM MobiSys*, 2021, pp. 54–66.

- [13] R. Presotto, G. Civitaresse, and C. Bettini, "SemFed: Semantic-based federated learning for heterogeneous activity recognition," in *Proc. IEEE PerCom*, 2022, pp. 1–10.
- [14] Y. Zhao, M. Li, L. Lai, N. Suda, D. Civin, and V. Chandra, "Federated learning with non-IID data," *arXiv preprint arXiv:1806.00582*, 2018.
- [15] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. AISTATS*, Fort Lauderdale, FL, 2017, pp. 1273–1282.
- [16] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, "Federated optimization in heterogeneous networks," in *Proc. MLSys*, 2020, pp. 429–450.
- [17] D. Li and J. Wang, "FedMD: Heterogeneous federated learning via model distillation," in *NeurIPS Workshop on Federated Learning*, Vancouver, 2019.
- [18] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," in *NeurIPS Deep Learning Workshop*, Montreal, 2015.
- [19] T. Shen, J. Zhang, X. Jia, F. Zhang, G. Huang, P. Zhou, K. Cheng, and F. Wu, "Federated mutual learning," *arXiv preprint arXiv:2006.16765*, 2020.
- [20] T. Lin, L. Kong, S. U. Stich, and M. Jaggi, "Ensemble distillation for robust model fusion in federated learning," in *Proc. NeurIPS*, 2020, pp. 2351–2363.
- [21] G. Gad, "HomeSenseKD: Heterogeneous federated knowledge distillation for CSI-based indoor sensing," *GitHub*, 2024. [Online]. Available: <https://github.com/gadm21/HomeSenseKD>